Nonuniqueness for degenerate Hermitian Monge–Ampère equations

Abstract

We construct on $\mathbf{P}^3$ a Hermitian form ω, a smooth function $f \geq 0$, $f \not\equiv 0$, and two distinct smooth normalized ω-plurisubharmonic functions φ, ψ with $(\omega + dd^c\varphi)^3 = f\omega_X^3 = (\omega + dd^c\psi)^3$. This answers negatively the uniqueness question for degenerate Hermitian Monge–Ampère equations raised by Boucksom–Guedj–Lu.

Keywords. Complex Monge–Ampère equation, Hermitian manifold, degenerate density, uniqueness.

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1 Introduction

Let X be a compact complex manifold of dimension n, let ω_X be a Hermitian form, and let θ be a smooth real $(1,1)$-form, not necessarily closed. Boucksom, Guedj and Lu proved that if θ is big and $0 \leq f \in L^p(X)$ for some $p > 1$, with $f \not\equiv 0$, then there exist a normalized θ-psh function u with minimal singularities and a constant $c > 0$ satisfying

$$(\theta + dd^c u)^n = cf\omega_X^n, \tag{1.1}$$

and that c is uniquely determined [BGL25, Theorem D]. Uniqueness of u is stated there to be open in the Hermitian setting, even when θ is a Hermitian form; Kołodziej and Nguyen proved it when the density is bounded away from zero [KN19], and for smooth positive densities and smooth solutions it follows from the maximum principle, as in the work of Tosatti and Weinkove [TW10]. When θ is Kähler, uniqueness holds for all densities [Din09].

We show that the positivity assumption on the density cannot be dropped for a nonclosed Hermitian form, even on projective space.

Theorem A. *Let $X = \mathbf{P}^3$ with its Fubini–Study form ω_X. There exist a Hermitian form $\omega > 0$ with $d\omega \neq 0$, a function $f \in C^\infty(X)$ with $f \geq 0$, $f \not\equiv 0$, vanishing on a nonempty open set, and two distinct functions*

$$\varphi, \psi \in C^\infty(X) \cap \mathrm{PSH}(X, \omega)$$

such that

$$\sup_X \varphi = \sup_X \psi = 0, \qquad (\omega + dd^c\varphi)^3 = f\omega_X^3 = (\omega + dd^c\psi)^3.$$

Both potentials have minimal singularities.

The mechanism is elementary. On a dense open subset of $\mathbf{P}^3$ there are three rank-one semipositive $(1,1)$-forms P, Q, R whose wedge product is positive, so that $(xP + yQ + zR)^3 = 6xyz\, P \wedge Q \wedge R$ for $x, y, z \geq 0$. All forms in the construction are combinations of P, Q, R with coefficients depending on a single real function r, and the Monge–Ampère measures are read off from products of three coefficients. We produce a semipositive form ω_0 and a nonconstant function w such that $\omega_0 \pm \frac{1}{2}\mathrm{dd^c}w$ are both semipositive with vanishing cube on the band where w is nonconstant. The form ω_0 degenerates along two real hypersurfaces, and a small correction $\mathrm{dd^c}\rho$ makes $\omega = \omega_0 + \mathrm{dd^c}\rho$ positive; the potentials $\mp\frac{1}{2}w - \rho$ then have the same Monge–Ampère measure with respect to ω. Since uniqueness holds when ω is closed, ω is necessarily nonclosed.

We use the conventions

$$d^c = i(\bar\partial - \partial), \qquad \mathrm{dd^c} = 2i\partial\bar\partial,$$

and $\mathrm{PSH}(X, \omega)$ denotes the set of ω-psh functions, that is, upper semicontinuous u with $\omega + \mathrm{dd^c}u \geq 0$.

2 A rank-one frame on projective three-space

Write $X = \mathbf{P}(\mathbf{C}^2 \oplus \mathbf{C}^2) = \mathbf{P}^3$ with homogeneous coordinates $[A : B]$, $A, B \in \mathbf{C}^2$, and let

$$U := \{A \neq 0,\ B \neq 0\}, \qquad r := \log \frac{|A|^2}{|B|^2} \in C^\infty(U).$$

On U put

$$P := \mathrm{dd^c}\log|A|^2, \qquad Q := \mathrm{dd^c}\log|B|^2, \qquad R := dr \wedge d^c r = 2i\,\partial r \wedge \bar\partial r.$$

Thus P and Q are the pullbacks of twice the Fubini–Study form of $\mathbf{P}^1$ under $[A : B] \mapsto [A]$ and $[A : B] \mapsto [B]$. Let

$$\sigma(r) := \frac{e^r}{1+e^r}, \qquad \sigma'(r) = \frac{e^r}{(1+e^r)^2},$$

and let $\omega_X := \mathrm{dd^c}\log(|A|^2 + |B|^2)$ be the Fubini–Study form of X in our normalization.

Lemma 2.1. *The forms P, Q, R are semipositive of rank one, and*

$$P^2 = Q^2 = R^2 = 0, \qquad P \wedge Q \wedge R > 0 \quad \text{on } U. \tag{2.1}$$

For every $G \in C^\infty(\mathbf{R})$,

$$\mathrm{dd^c}\, G(r) = G'(r)\,(P - Q) + G''(r)\,R. \tag{2.2}$$

In particular, for $x, y, z \geq 0$,

$$(xP + yQ + zR)^3 = 6xyz\, P \wedge Q \wedge R, \tag{2.3}$$

so $xP + yQ + zR$ is semipositive, and positive exactly where $xyz > 0$; and

$$\omega_X = \sigma(r)P + \sigma(-r)Q + \sigma'(r)R \qquad \text{on } U. \tag{2.4}$$

Proof. Rank one and semipositivity are clear from the definitions. Since $\mathrm{dd^c}r = P - Q$, the chain rule gives $\mathrm{dd^c}G(r) = G'(r)\mathrm{dd^c}r + G''(r)\,dr \wedge d^c r$, which is (2.2); applied to $G(r) = \log(1 + e^r)$, for which $\log(|A|^2 + |B|^2) = \log|B|^2 + G(r)$, it gives (2.4). For (2.1) use the local coordinates $A = (1, a)$, $B = \tau(1, b)$ on the corresponding chart of U (the other charts are analogous). Then P and Q are positive multiples of $i\,da \wedge d\bar a$ and $i\,db \wedge d\bar b$, while

$$\partial r = \frac{\bar a\, da}{1 + |a|^2} - \frac{d\tau}{\tau} - \frac{\bar b\, db}{1 + |b|^2}$$

has a nonzero $d\tau$-component, so the three rank-one directions are independent; explicitly,

$$P \wedge Q \wedge R = \frac{8\,(i\,da \wedge d\bar a) \wedge (i\,db \wedge d\bar b) \wedge (i\,d\tau \wedge d\bar\tau)}{(1 + |a|^2)^2(1 + |b|^2)^2|\tau|^2} > 0.$$

Finally (2.3) follows from (2.1) by expanding the cube. □

A function of the form $G(r)$ with G constant on $(-\infty,-2]$ and on $[2,\infty)$ extends smoothly to X, being constant near $\{A=0\}$ and $\{B=0\}$. Likewise, by (2.4), if x,y,z are smooth functions on $\mathbf{R}$ with

$$(x,y,z)=(\sigma,\sigma(-\,\cdot\,),\sigma') \quad \text{on } (-\infty,-2]\cup[2,\infty), \tag{2.5}$$

then $x(r)P+y(r)Q+z(r)R$ extends smoothly to X, being equal to ω_X near $\{A=0\}\cup\{B=0\}$. We shall use this tacitly.

3 The construction

One-variable data

Choose an even function $\beta\in C^\infty(\mathbf{R})$ with $\operatorname{supp}\beta=[-1,1]$, $\beta>0$ and $\beta'>0$ on $(-1,0)$, normalized by $\int_{-1}^{1}\beta=2$; for instance a multiple of $\exp(-1/(1-r^2))$ on $(-1,1)$, extended by zero. Let

$$F(r)=-1+\int_{-1}^{r}\beta(t)\,dt \quad (-1<r<1), \qquad F=-1 \text{ on } (-\infty,-1], \quad F=1 \text{ on } [1,\infty),$$

a smooth odd function, and put

$$p:=\tfrac12 F'=\tfrac12\beta, \qquad q:=\tfrac12 F''=\tfrac12\beta'. \tag{3.1}$$

Then p is even and positive on $(-1,1)$, q is odd, $q>0$ on $(-1,0)$ and $q<0$ on $(0,1)$, and p,q vanish to infinite order at ±1.

Choose a smooth function $a\geq 0$ on $\mathbf{R}$ with $a=1$ on $(-\infty,-\frac23]$ and $a=0$ on $[-\frac13,\infty)$, and put $b(r):=a(-r)$. Choose an even $\chi\in C_c^\infty((-\frac13,\frac13))$ with $\chi\geq 0$ and $\chi(0)>0$, and put

$$z_0:=(q^2+\chi^2)^{1/2} \qquad \text{on } [-1,1]. \tag{3.2}$$

This is smooth: $q^2+\chi^2>0$ on $(-1,1)$, because q vanishes there only at 0 and $\chi(0)>0$; near ±1 one has $\chi=0$ and $z_0=|q|$ with q of constant sign. Moreover

$$z_0\geq|q|, \qquad z_0=q \text{ where } a\neq 0, \qquad z_0=-q \text{ where } b\neq 0, \qquad z_0>0 \text{ on } (-1,1), \tag{3.3}$$

and z_0 vanishes to infinite order at ±1.

Finally choose smooth functions λ,s on $(-\infty,-1]$ with $0\leq\lambda\leq1$, $\lambda=0$ on $(-\infty,-2]$, $\lambda=1$ on $[-\frac32,-1]$; $s>0$ on $(-\infty,-1)$, $s=1$ on $(-\infty,-\frac32]$, and s vanishing to infinite order at -1.

The coefficient functions

Define $x,y,z\colon\mathbf{R}\to[0,\infty)$ by

$$\begin{aligned}
&x=(1-\lambda)\sigma+\lambda, && y=(1-\lambda)\sigma(-\,\cdot\,)+\lambda s, && z=(1-\lambda)\sigma'+\lambda s \qquad \text{on } (-\infty,-1],\\
&x=p+a, && y=p+b, && z=z_0 \qquad \text{on } [-1,1],\\
&x(r)=y(-r), && y(r)=x(-r), && z(r)=z(-r) \qquad \text{on } [1,\infty).
\end{aligned}$$

These functions are smooth on $\mathbf{R}$: the two definitions at $r=-1$ have values $(1,0,0)$ and differ by functions vanishing to infinite order there (p, q, z_0 and s are flat at -1, and $a=1$, $b=0$ near -1), and the definitions on $[-1,1]$ are symmetric under $r\mapsto -r$ since p, z_0 are even and $b(r)=a(-r)$. They satisfy (2.5), and

$$x,y,z>0 \text{ on } \mathbf{R}\setminus\{\pm1\}, \qquad (x,y,z)(-1)=(1,0,0), \qquad (x,y,z)(1)=(0,1,0). \tag{3.4}$$

Indeed, on $(-\infty,-1)$ all three are convex combinations, with weights $1-\lambda, \lambda$, of positive numbers, except that $x=1$ and $y=z=s>0$ on $[-\frac{3}{2},-1)$; on $(-1,1)$ we have $x \geq p>0$, $y \geq p>0$, $z=z_0>0$.

Now set, on U and hence on X,

$$\omega_0 := x(r)P + y(r)Q + z(r)R, \qquad w := F(r), \qquad H := \tfrac{1}{2}\mathrm{dd^c} w = p(r)(P-Q) + q(r)R, \tag{3.5}$$

where the last identity is (2.2). By Lemma 2.1 and (3.4), ω_0 is a smooth semipositive form on X, positive outside the two real hypersurfaces

$$M_- := \{r=-1\}, \qquad M_+ := \{r=1\}, \tag{3.6}$$

and w is a smooth function on X which is nonconstant, equal to -1 on $\{r \leq -1\}$ and to 1 on $\{r \geq 1\}$.

Proposition 3.1. *The forms*

$$A_0 := \omega_0 - H, \qquad B_0 := \omega_0 + H$$

are smooth semipositive forms on X, equal to ω_0 on $\{|r| \geq 1\}$, and

$$A_0^3 = B_0^3 = 6xyz\,\mathbf{1}_{\{|r|\geq 1\}}\, P \wedge Q \wedge R.$$

In particular $A_0^3 = B_0^3$ is a smooth semipositive volume form, not identically zero, vanishing exactly on $\{|r| \leq 1\}$.

Proof. Outside $\{|r|<1\}$ we have $p=q=0$, so $H=0$ and $A_0=B_0=\omega_0$, whose cube is $6xyz\, P\wedge Q\wedge R$ by (2.3); it is positive for $|r|>1$ and vanishes on $M_\pm$ by (3.4). On $\{|r|<1\}$,

$$\begin{aligned} A_0 &= aP + (2p+b)Q + (z_0-q)R, \\ B_0 &= (2p+a)P + bQ + (z_0+q)R, \end{aligned} \tag{3.7}$$

with all coefficients nonnegative by (3.3), so $A_0, B_0 \geq 0$, and by (2.3)

$$A_0^3 = 6a(2p+b)(z_0-q)\, P\wedge Q\wedge R, \qquad B_0^3 = 6(2p+a)b(z_0+q)\, P\wedge Q\wedge R.$$

By (3.3), $a(z_0-q)=0$ and $b(z_0+q)=0$ on $[-1,1]$, so both cubes vanish identically on $\{|r|<1\}$. ☐

4 Passage to a Hermitian background

The form ω_0 is degenerate along $M_\pm$, and this cannot be avoided within the ansatz: semipositivity of A_0 and B_0 on the band requires $x \geq p$, $y \geq p$ and $z \geq |q|$, so $x+p>0$ and $z+q>0$ on $(-1,0)$; then the vanishing of B_0^3 forces $y=p$ on $(-1,0)$, and the vanishing of A_0^3 forces $x=p$ or $z=q$ at each point of $(-1,0)$. Letting $r\to -1$ shows that y and at least one of x,z vanish at $r=-1$, so ω_0 has rank at most one on M_-. The following change of representative removes the degeneracy without changing the two Monge–Ampère measures.

Proposition 4.1. *Let $\kappa \in C_c^\infty((-2,2))$ with $0 \leq \kappa \leq 1$ and $\kappa = 1$ on $[-\frac{3}{2},\frac{3}{2}]$, and put*

$$\rho := \varepsilon\,(r^2-1)\,\kappa(r) \in C^\infty(X). \tag{4.1}$$

For all sufficiently small $\varepsilon>0$, the form

$$\omega := \omega_0 + \mathrm{dd^c}\rho$$

is positive on X. Moreover $d\omega \neq 0$.

Proof. By (2.2), on U,

$$\omega = (x+\rho')P + (y-\rho')Q + (z+\rho'')R,$$

with ρ', ρ'' the derivatives of ρ as a function of r; near $\{A=0\} \cup \{B=0\}$ we have $\rho = 0$ and $\omega = \omega_0 = \omega_X > 0$. By Lemma 2.1 it suffices to show that the three coefficients are positive on $\mathbf{R}$ for small ε.

On $[-\frac{3}{2}, \frac{3}{2}]$ we have $\rho' = 2\varepsilon r$ and $\rho'' = 2\varepsilon$. Here $z + 2\varepsilon > 0$. For $r \in [0, \frac{3}{2}]$, $x + 2\varepsilon r \geq x$ with equality only at $r = 0$, where $x(0) = p(0) > 0$; for $r \in [-\frac{3}{2}, 0]$, $x + 2\varepsilon r \geq m - 3\varepsilon$ with $m := \min_{[-3/2,0]} x > 0$ by (3.4). Thus $x + \rho' > 0$ on $[-\frac{3}{2}, \frac{3}{2}]$ for $\varepsilon < m/3$, and the symmetric argument gives $y - \rho' > 0$ there. On the compact set $\frac{3}{2} \leq |r| \leq 2$ the functions x, y, z are bounded below by a positive constant m' by (3.4), while $|\rho'|, |\rho''| \leq \varepsilon M$ for a constant M depending only on κ; so all coefficients are positive for $\varepsilon < m'/M$. For $|r| \geq 2$, $\rho = 0$ and $(x, y, z) = (\sigma, \sigma(-\,\cdot\,), \sigma') > 0$.

For nonclosedness, $d\omega = d\omega_0$ since $dd\mathrm{d^c}\rho = 0$. As $R = dr \wedge d^c r$ and $\mathrm{dd^c} r = P - Q$, we have $dR = -dr \wedge (P - Q)$ and $dr \wedge R = 0$, so

$$d\omega_0 = dr \wedge \big((x' - z)P + (y' + z)Q\big).$$

At $r = 0$ we have $x' = p' + a' = q(0) = 0$, likewise $y' = 0$, and $z(0) = \chi(0) > 0$, so $d\omega_0 = z(0)\, dr \wedge (Q - P) \neq 0$, because $dr \wedge P$ and $dr \wedge Q$ are linearly independent by the independence in Lemma 2.1. □

5 Proof of the main theorem

Proof of Theorem A. Let $\omega = \omega_0 + \mathrm{dd^c}\rho$ be the Hermitian form of Proposition 4.1, and define

$$\widetilde{\varphi} := -\tfrac{1}{2}w - \rho, \qquad \widetilde{\psi} := \tfrac{1}{2}w - \rho.$$

By (3.5),

$$\omega + \mathrm{dd^c}\widetilde{\varphi} = \omega_0 - H = A_0, \qquad \omega + \mathrm{dd^c}\widetilde{\psi} = \omega_0 + H = B_0, \tag{5.1}$$

which are semipositive by Proposition 3.1. Hence $\varphi := \widetilde{\varphi} - \sup_X \widetilde{\varphi}$ and $\psi := \widetilde{\psi} - \sup_X \widetilde{\psi}$ are smooth normalized ω-psh functions, and (5.1) still holds for them. They are distinct, since $\psi - \varphi = w + \mathrm{const}$ is nonconstant. Put

$$f := \frac{A_0^3}{\omega_X^3} = \frac{B_0^3}{\omega_X^3}.$$

By Proposition 3.1, f is a smooth nonnegative function, not identically zero, vanishing on the nonempty open set $\{|r| < 1\}$, and

$$(\omega + \mathrm{dd^c}\varphi)^3 = f\omega_X^3 = (\omega + \mathrm{dd^c}\psi)^3.$$

Finally, since $\omega > 0$, the constant 0 is ω-psh, so the envelope $V_\omega = \sup\{u \in \mathrm{PSH}(X, \omega) : u \leq 0\}$ is identically zero, and every bounded ω-psh function has minimal singularities; this applies to φ and ψ. Proposition 4.1 gives $d\omega \neq 0$. □

Remark 5.1. Both solutions correspond to the same constant $c = 1$ in (1.1), in accordance with the uniqueness of c. The manifold $\mathbf{P}^3$ is Kähler, so the example is not related to the failure of the bounded mass property, and the two solutions are smooth, so the failure of uniqueness is not a regularity phenomenon. It is instructive to see why the maximum principle, which gives uniqueness for smooth solutions with positive density [TW10], yields nothing here. Writing $A_0^3 - B_0^3 = -\mathrm{dd^c}w \wedge T$ with $T = A_0^2 + A_0 \wedge B_0 + B_0^2$, the function w solves the linear equation $T \wedge \mathrm{dd^c}w = 0$, which is elliptic where T is positive. Here T is positive on $X \setminus (M_- \cup M_+)$, but on

$M_\pm$ it equals $3\omega_0^2 = 0$ because ω_0 has rank one there; and $M_- \cup M_+$ is precisely the boundary of the band on which w is nonconstant, so the strong maximum principle cannot propagate the constancy of w into the band. The essential feature of the example is thus the vanishing of the density on an open set; we do not know whether uniqueness holds when the zero set of the density has empty interior.

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